VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

(An Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering)

Department of Computer Engineering

Project Report on

Genotype Inference from Leaf Phenotype Using Venation Patterns and Deep Learning for rice crop varieties

In partial fulfillment of the Third Year (Semester–VI), Bachelor of Technology (B.Tech.) Degree in Computer Engineering at the University of Mumbai

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(2025-26)

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CERTIFICATE of Approval

This is to certify that ***Shivam Chaugule*** of Third Year Computer Engineering studying under the University of Mumbai has satisfactorily presented the project on **"Genotype Inference from Leaf Phenotype Using Venation Patterns and Deep Learning for rice crop varieties"** as a part of the coursework of CAPSTONE PROJECT - I for Semester-VI under the guidance of ***Dr. Priya R. L.*** in the year 2025-26.

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We wish to express our profound thanks to all those who helped us in gathering information about the project. Our families too have provided moral support and encouragement several times.

Computer Engineering Department

COURSE OUTCOMES FOR CAPSTONE PROJECT

Learners will be to:-

Course Outcomes	Description of the Course Outcomes
CO1	Identify and define a real-world problem through systematic requirement analysis and investigation.
CO2	Conduct a thorough literature review and benchmarking to propose innovative solutions.
CO3	Develop a structured project plan with well-defined milestones, timelines, and risk mitigation strategies.
CO4	Design and implement a working prototype using appropriate engineering tools and methods.
CO5	Analyze experimental results, validate the solution, and optimize performance based on data.
CO6	Effectively document and present the project outcomes, demonstrating professional ethics, teamwork, and project management skills.

ABSTRACT

Leaves are among the most accessible and informative plant organs, as they exhibit a wide range of morphological characteristics such as shape, size, texture, and venation patterns. Among these features, leaf venation represents the vascular network responsible for the transport of water, nutrients, and photosynthetic products within the plant. This venation structure is not only crucial for plant physiology but is also a stable phenotypic trait that is largely governed by genetic regulation. Unlike other visual features that may vary significantly due to environmental conditions, venation patterns tend to remain consistent, making them a reliable indicator for studying plant characteristics and underlying genetic information.

Traditionally, genotypic analysis of plants involves molecular techniques such as DNA sequencing, PCR, and marker-based studies. While these methods provide highly accurate genetic information, they are expensive, time-consuming, and require specialized laboratory infrastructure. As a result, they are not practical for large-scale applications, especially in agricultural settings where rapid and cost-effective solutions are needed.

To overcome these limitations, phenotype-based analysis has emerged as a promising alternative. By utilizing observable traits such as leaf morphology and venation, it becomes possible to infer genotype-associated information in a non-invasive manner. With the advancement of computer vision and deep learning, it is now feasible to automatically extract complex phenotypic patterns from images and establish meaningful relationships with genetic traits.

In this project, a deep learning-based approach is proposed to infer genotype-associated information from the leaf phenotypes of rice plants, with a primary focus on venation patterns. Convolutional Neural Networks (CNNs) are employed to automatically learn hierarchical features from leaf images, capturing intricate details such as vein density, branching structure, and overall vascular topology. These extracted features are then used to map phenotypic characteristics to genotype-related classes, such as rice varieties or genetic groups.

The proposed system aims to provide a scalable, non-invasive, and efficient solution for plant phenotyping and genotype inference. By reducing reliance on laboratory-based genetic analysis and leveraging image-based techniques, the system has potential applications in agriculture, crop improvement, and plant research, particularly for large-scale analysis of rice varieties.

INDEX

Chapter No.	Title	Page No.
1	**Introduction**	**9**
1.1	Introduction to the project	9
1.2	Motivation for the project	9
1.3	Drawback of the existing system	9
1.4	Problem Definition	10
1.5	Relevance of the Project	11
1.6	Methodology used	**11**
2	**Literature Survey**	14
2.1	Research Papers	15
2.2	Books / Articles / Newspaper referred	15
2.3	Interaction with domain experts	15
2.4	Patent Search	18
3	**Requirement of Proposed System**	**20**
3.1	Functional Requirements	20
3.2	Non-Functional Requirements	22
3.3	Constraints	24
3.4	Hardware & Software Requirements	24
3.5	Techniques utilized till date for the proposed system	25
3.6	Tools utilized till date for the proposed system	25
3.7	SDGs mapped with the proposed project and justification	25
4	**Proposed Design**	**27**

4.1	Block diagram representation of the proposed system with explanation	27
4.2	Modular diagram representation of the proposed system with explanation	28
4.3	Design of the proposed system – Data Flow / Use Case Diagram; Flowchart; State Transition / Activity Diagram; ER Diagram; Screenshot of implementation	29
4.4	Algorithms utilized in the existing systems	30
4.5	Project scheduling and tracking using Timeline / Gantt Chart	37
5	**Implementation Results and Discussions**	**38**
5.1	Data Collection / Acquisition	38
5.2	Data Preprocessing Techniques	38
5.3	Experimental Setup	41
5.4	Evaluation Metrics	41
5.5	Gap analysis of existing systems	44
6	**Plan of Action for the Next Semester**	**45**
6.1	Work done till date	45
6.2	Plan of action for Project II	45
7	**Conclusion**	**48**
8	**References (IEEE format)**	**49**
9	**Appendix**	**51**
9.1	List of Figures	51
9.2	List of Tables	51
9.3	Paper Publications – Acceptance mail; Published paper; Plagiarism report; Certificate	52
9.4	Xerox of Project Review 1 & 2 Evaluation sheet	53

1. Introduction

1.1 Introduction to the Project :

Understanding the relationship between genotype and phenotype is an important area in plant science. Genotype refers to the genetic makeup of a plant, while phenotype includes observable traits like leaf shape, texture, and venation.

This project focuses on using leaf venation patterns of rice plants to infer genotype-related information. Venation is a stable and biologically significant feature, which makes it useful for studying underlying genetic traits.

Instead of using lab-based genetic methods, the project uses image processing and deep learning to extract features from leaf images and relate them to genotype. The idea is to create a non-invasive and scalable system for plant analysis.

The shift to rice plants was made because rice has well-established genomic datasets and research support, making it more suitable for this study.

1.2 Motivation for the Project

Traditional methods of genotype identification, such as DNA sequencing, are accurate but expensive and time-consuming. They also require laboratory setups, which are not always practical for large-scale agricultural use.

Rice, being a major crop, has a large amount of genomic data available, including studies involving thousands of varieties . However, using this data directly is complex.

At the same time, leaf features like venation are easily accessible and contain useful biological information. With the help of deep learning, it becomes possible to analyze these patterns automatically.

So, the main motivation is to develop a system that can use simple leaf images to estimate genetic information, making the process faster and more practical.

1.3 Drawbacks of Existing Systems

Current systems have a few limitations:

- Most genotype analysis methods depend on lab-based techniques, which are costly and not scalable.

- Image-based models mainly focus on plant classification or disease detection, not genotype inference.
- Many approaches rely on color and texture, which can change with environmental conditions.
- Venation patterns, which are more genetically relevant, are often not fully utilized.
- Genomic studies provide detailed data, but they are not integrated with image-based systems .

1.4 Problem Definition

The primary objective of this project is to develop a system capable of inferring genotype-related information from rice leaf images by analyzing phenotypic characteristics, particularly leaf venation patterns, using deep learning techniques. The system aims to establish a meaningful relationship between observable traits (phenotype) and the underlying genetic composition (genotype) of rice plants, thereby enabling a non-invasive approach to genotype analysis.

One of the key challenges in this problem is that genotype information is not directly visible and is traditionally obtained through molecular methods such as DNA sequencing, which are expensive, time-consuming, and require specialized laboratory infrastructure. In contrast, phenotype-based features such as leaf shape, texture, and especially venation patterns are easily accessible but only indirectly reflect genetic information. Bridging this gap between phenotype and genotype requires sophisticated modeling techniques capable of capturing subtle patterns and correlations.

Another major challenge lies in accurately extracting and representing venation structures from leaf images, as these features must be robust to variations in lighting, orientation, and image quality. Furthermore, the relationship between phenotypic traits and genotype is complex and often influenced by multiple factors, including environmental conditions. Therefore, the system must effectively learn high-level representations from image data and integrate them with genomic knowledge using deep learning and data-driven approaches.

Overall, the problem involves designing an efficient, scalable, and accurate system that can map visual leaf features to genotype-associated information, providing a practical alternative to traditional genetic analysis methods and supporting applications in agriculture, crop improvement, and plant research.

1.5 Relevance of the Project

This project is useful in both agriculture and research.

- Helps in identifying rice varieties
- Supports crop improvement and breeding programs
- Provides a faster and non-invasive alternative to traditional methods

1.6 Methodology Used

The project follows a step-by-step approach:

1. Data Collection

- Collect rice leaf images
- Use genomic datasets from sources like SRA and SNP-Seek

Table 1: Dataset Collected

Sr. No.	Dataset Name	Link	Image Type / Description	Category
1	Rice Image Dataset (Arborio, Basmati, Ipsala, Jasmine, Karacadag)	https://www.kaggle.com/datasets/muratkokludataset/rice-image-dataset	75,000 individual grain images (15,000 per class). White-background macro shots of single grains at 250×250 px. Five globally grown varieties with distinct shape, size, and texture differences.	**Grain Variety**
2	Aruzz22.5K — 20 Bangladesh Rice Varieties (Mendeley)	https://data.mendeley.com/datasets/3mn9843tz2/4	4,730 original + 23,650 augmented JPG images (512×512 px) of 20 Bangladeshi varieties such as Bashmoti, Chinigura, Lal Aush, and Paijam. Captured with an iPhone 11 at 5× zoom.	**Grain Variety**
3	38 Rice Varieties — BINA/BRRI (Mendeley)	https://www.sciencedirect.com/science/article/pii/S2352340924010710	19,000 original + 57,000 augmented images of 38 Bangladeshi rice varieties (BD33, BD30, BD56, etc.) shot under dual microscope cameras. Covers grain color, shape, texture, and size.	**Grain Variety**
4	PRBD — Microscopic Processed Rice (10 Bangladesh Varieties)	https://www.sciencedirect.com/science/article/pii/S2352340925007802	~2,010 original + 8,010 augmented high-resolution microscopic images of 10 milled rice types. Focuses on post-milling grain structure, polishing quality, and nutritional loss indicators.	**Grain Variety**

5	Rice Type Classification (Kaggle)	https://www.kaggle.com/datasets/mssmartypants/rice-type-classification	Compact grain image dataset with clean class labels for rice type classification. Suitable for quick prototyping and benchmarking lightweight models.	**Grain Variety**
6	CVRP — Rice Panicle & Cultivar Dataset (281 varieties, Nanjing)	https://www.sciencedirect.com/science/article/pii/S2643651525000317	High-resolution field and indoor images of 231 landraces + 50 modern rice cultivars. Includes both natural-pose and manually-unfolded panicle shots for grain counting and cultivar ID.	**Panicle / Plant**
7	Rice Plant Image Dataset (Mendeley)	https://data.mendeley.com/datasets/8ybmgnm3p3/2	General-purpose rice plant images covering form, color, texture, and physiological aspects of the whole plant. Useful for broad agricultural computer vision research.	**Plant / General**
8	Rice Leaf Diseases — UCI Repository (Bacterial Blight, Brown Spot, Leaf Smut)	https://archive.ics.uci.edu/ml/datasets/Rice+Leaf+Diseases	120 JPG images (40 per class) of diseased rice leaves covering 3 disease types: Bacterial Leaf Blight, Brown Spot, and Leaf Smut. Classic benchmark dataset for leaf disease classification.	**Leaf Disease**
9	Rice Leaf Diseases (Kaggle — vbookshelf)	https://www.kaggle.com/datasets/vbookshelf/rice-leaf-diseases	Leaf-level disease images across the same 3 disease categories as the UCI set. Widely used for CNN and transfer learning benchmarks. Well-structured train/test split available.	**Leaf Disease**
10	Rice Leaf Disease Image (Kaggle — nirmalsankalana)	https://www.kaggle.com/datasets/nirmalsankalana/rice-leaf-disease-image	Larger multi-class leaf disease dataset including healthy leaf images. Broader category coverage makes it suitable for multi-label classification and real-world generalization testing.	**Leaf Disease**

2. Image Preprocessing

- Resize images to a standard resolution to ensure uniform input
- Normalize pixel values for consistent model training
- Remove background and noise to isolate the leaf region
- Apply enhancement techniques (e.g., CLAHE) to improve visibility of venation patterns

3. Feature Extraction

- Extract venation features such as vein density and branching patterns
- Capture texture features representing surface variations
- Derive shape and morphological features like contour and structure

4. Model Training

- Use CNN-based models such as ResNet or EfficientNet
- Train the model on processed leaf images
- Enable the model to learn complex patterns and relationships in the data

5. Feature Combination

- Combine features from different sources (venation, texture, shape)
- Create a unified representation of the leaf phenotype
- Improve prediction accuracy by capturing multiple aspects of the data

6. Genotype Mapping

- Link extracted phenotypic features with genotype-related information
- Use genomic datasets (e.g., SRA, SNP-Seek)
- Apply data-driven approaches to establish phenotype–genotype relationships

7. Evaluation

- Evaluate model performance using metrics such as accuracy, precision, and F1-score
- Use venation-specific metrics (e.g., edge density, vein coverage)
- Validate results to ensure reliability and consistency

2. Literature Survey

Table 2:Literature review

Sr. No.	Research Paper Title	Author(s)	Year	DOI	Link to Research Paper	Database Used
1	A Pangenome reference of wild and cultivated rice	Guo D, Li Y, Lu H et al.	2025	10.1038/s41586-025-08883-6	https://www.nature.com/articles/s41586-025-08883-6	NCBI SRA / Zenodo
2	The rice genome annotation project: an updated database for mining the rice genome	Hamilton JP, Li C, Buell CR	2025	10.1093/nar/gkae1061	https://academic.oup.com/nar/article/53/D1/D1614/7903367	Rice Genome Annotation Project (RGAP)
3	A gapless genome assembly of a Japonica variety 'BD8' provides insights into rice salt tolerance	Zhang et al.	2025	10.3389/fpls.2025.1713117	https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2025.1713117/full	NCBI / RAP-DB
4	The Oryza australiensis genome reveals potential as a source of genes for rice improvement	Furtado A et al.	2025	10.1038/s41598-025-13310-x	https://www.nature.com/articles/s41598-025-13310-x	NCBI GenBank
5	An improved haplotype resolved genome reveals more rice genes	Abdullah M, Furtado A, Masouleh AK	2024	10.48130/tp-0024-0007	https://www.maxapress.com/article/doi/10.48130/tp-0024-0007	NCBI GenBank / OmicsBox
6	Progress in rice breeding based on genomic research	Yan X, Wu H,	2024	10.3390/genes15050564	https://www.mdpi.com/2073-4425/15/5/564	NCBI / 3000 Rice Genomes

		Yu S et al.				Project (SNP-Seek)
7	Genome-wide and pan-genomic analysis reveals rich variants of NBS-LRR genes in a newly developed wild rice line from Oryza alta Swallen	Shivute FN, Zhong Y, Wu J et al.	2024	10.3389/fpls.2024.1345708	https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2024.1345708/full	NCBI SRA / MSU7 Reference (rice.uga.edu)
8	Genome evolution and diversity of wild and cultivated rice species	Zhou Y et al.	2024	10.1038/s41467-024-54427-3	https://www.nature.com/articles/s41467-024-54427-3	NCBI SRA
9	A complete assembly of the rice Nipponbare reference genome (T2T-NIP)	Shang L, Li X, He H et al.	2023	10.1016/j.molp.2023.08.005	https://www.cell.com/molecular-plant/fulltext/S1674-2052(23)00219-8	NCBI GenBank / RAP-DB
10	A complete telomere-to-telomere assembly provides new reference genome for rice	Huang X	2023	10.1016/j.molp.2023.08.007	https://www.cell.com/molecular-plant/fulltext/S1674-2052(23)00223-X	NCBI GenBank
11	Whole-genome sequencing and RNA-Seq reveal differences in genetic mechanism for flowering response between weedy rice and cultivated rice	Lim SH, Kim DY, Lee SM et al.	2022	10.3390/ijms23031608	https://www.mdpi.com/1422-0067/23/3/1608	NCBI SRA / 3K Rice Genome (RG) Dataset (SNP-Seek)

12	Pan-genome analysis highlights the extent of genomic variation in cultivated and wild rice	Zhao Q, Feng Q, Lu H et al.	2022	10.1038/s41588-018-0041-z	https://www.nature.com/articles/s41588-018-0041-z	NCBI SRA / Rice Pan-genome DB
13	Long-read sequencing of 111 rice genomes reveals significantly larger pan-genomes	Zhang F et al.	2022	10.1101/gr.276015.121	https://genome.cshlp.org/content/32/5/853	NCBI SRA
14	Genomic variation in 3,010 diverse accessions of Asian cultivated rice	Wang W, Mauleon R, Hu Z et al.	2018	10.1038/s41586-018-0063-9	https://www.nature.com/articles/s41586-018-0063-9	NCBI SRA / SNP-Seek (IRRI)
15	A super pan-genomic landscape of rice	Shang L, Zhao F, Zhang C et al.	2022	10.1038/s41422-022-00685-z	https://www.nature.com/articles/s41422-022-00685-z	NCBI SRA / RiceSuperPIRdb
16	Genomic investigation of 18,421 lines reveals the genetic architecture of rice	Wei C et al.	2024	10.1126/science.adm8762	https://www.science.org/doi/10.1126/science.adm8762	NCBI SRA / SNP-Seek
17	The map-based sequence of the rice genome	International Rice Genome Sequencing Project (IRGSP); Sasaki T	2005	10.1038/nature03895	https://www.nature.com/articles/nature03895	NCBI GenBank / RAP-DB

18	Improvement of the Oryza sativa Nipponbare reference genome using next generation sequence and optical map data	Kawahara Y et al.	2013	10.1186/1939-8433-6-4	https://thericejournal.springeropen.com/articles/10.1186/1939-8433-6-4	NCBI GenBank / RAP-DB / RGAP
19	Whole-genome sequencing of Oryza brachyantha reveals mechanisms underlying Oryza genome evolution	Chen J et al.	2013	10.1038/ncomms2596	https://www.nature.com/articles/ncomms2596	NCBI GenBank
20	The genome sequence of African rice (Oryza glaberrima) and evidence for independent domestication	Wang M et al.	2014	10.1038/ng.3044	https://www.nature.com/articles/ng.3044	NCBI GenBank / Ensembl Gramene

2.4 Patent Search :

Patent 1 :

Title: Genes that control rice grain length, grain weight, yield and grain appearance quality and their applications

Patent Number: CN107384937B

Source / Link: https://patents.google.com/patent/CN107384937B/en

Inventors / Assignee Institute of Genetics and Developmental Biology, Chinese Academy of Sciences

Year: 2017

Description:

This patent describes the identification and application of the LGY3 gene, which plays a significant role in controlling important agronomic traits in rice such as grain length, grain weight, yield, and grain appearance quality. The invention highlights how specific allelic variations of the LGY3 gene can improve yield and enhance grain quality. It also introduces breeding methods using molecular markers to develop improved rice varieties by leveraging this gene.

Patent 2 :

Title: Method of Distinguishing Rice Varieties

Patent Number: AU2003242177

Source / Link: https://patentscope.wipo.int/search/en/detail.jsf?docId=AU181231150

Inventor(s): Motonobu Namura

Country / Office: Australia

Application Date: 10 June 2003

Publication Date: 11 September 2003

IPC Classification: C12Q 1/68 (Biochemical / genetic analysis methods)

Description:

This patent describes a method for distinguishing different rice varieties using molecular genetic analysis techniques. It utilizes DNA-based approaches such as RFLP (Restriction Fragment Length Polymorphism), CAPS, and SNP analysis to identify genetic differences between rice varieties. The method involves extracting genomic DNA, amplifying specific regions using PCR, and analyzing polymorphisms to differentiate closely related rice types. The invention aims to provide a reliable and accurate system for identifying rice varieties, particularly in cases where traditional visual or morphological methods are insufficient.

3. Requirement of Proposed System

3.1 Functional Requirements :

3.1.1 Input Module

The input module is responsible for acquiring data from the user in the form of rice leaf images. The system should allow users to upload images in standard formats such as JPG or PNG. It must validate the input to ensure that the uploaded file contains a valid leaf image and is suitable for processing. Additionally, the module may support optional input of genomic references or dataset identifiers, such as SRA IDs, to assist in genotype mapping. This module acts as the entry point of the system and ensures that only valid and usable data is passed to subsequent stages.

3.1.2 Preprocessing Module

The preprocessing module prepares the input images for further analysis by standardizing and enhancing them. It includes resizing images to a fixed resolution, normalizing pixel values, and removing background noise to isolate the leaf region. Techniques such as contrast enhancement (e.g., CLAHE) and filtering are applied to improve image clarity. This module ensures consistency across the dataset and reduces variations caused by environmental factors such as lighting and background conditions.

3.1.3 Venation Extraction Module

This module focuses on extracting the venation structure of the leaf, which is a key phenotypic feature. It applies image processing techniques such as edge detection, vesselness filtering, or segmentation algorithms to highlight the vein network. The goal is to generate a clear representation of vein patterns, including their density and connectivity. Since venation is strongly influenced by genetic factors, this module plays a critical role in linking phenotype to genotype.

3.1.4 Feature Extraction Module

The feature extraction module derives meaningful information from the processed images. It extracts multiple types of features, including venation features (such as vein density and branching patterns), texture features (such as intensity variations and surface patterns), and morphological features (such as leaf shape, contour, and symmetry). These features provide a comprehensive representation of the leaf phenotype and are used as input for the learning model.

3.1.5 Deep Learning Module

This module is responsible for learning complex patterns from the extracted features using deep learning techniques. Models such as Convolutional Neural Networks (CNNs) or Transformer-based architectures are used to automatically identify important patterns within the data. The module supports both training and inference processes, enabling the system to generalize from training data and make predictions on new inputs. It eliminates the need for manual feature engineering and improves accuracy.

3.1.6 Feature Fusion Module

The feature fusion module combines features obtained from different sources, such as venation, texture, and shape, into a unified representation. This integration enhances the overall understanding of the leaf phenotype by considering multiple aspects simultaneously. By combining features, the system can capture both local and global patterns, leading to improved prediction performance.

3.1.7 Phenotype Prediction Module

This module uses the learned representations to predict observable traits of the rice plant. These may include classification of rice variety or prediction of specific phenotypic characteristics. The output of this module serves as an intermediate step before genotype mapping and reflects the visible characteristics derived from the leaf image.

3.1.8 Genotype Mapping Module

The genotype mapping module establishes a relationship between predicted phenotypic traits and underlying genetic information. It uses genomic datasets from sources such as NCBI SRA or SNP-Seek and applies concepts similar to Genome-Wide Association Studies (GWAS) to identify correlations between features and genetic markers. This module bridges the gap between phenotype and genotype and is a key component of the system.

3.1.9 Output Module

The output module presents the final results to the user in an understandable format. It displays predicted phenotype traits along with genotype-associated information, such as possible genetic markers or classifications. The output is designed to be clear and interpretable, enabling users to understand the results without requiring deep technical knowledge.

3.1.10 Evaluation Module

The evaluation module assesses the performance of the system. It calculates metrics such as accuracy, precision, recall, and F1-score, along with domain-specific metrics related to venation quality. This module ensures that the system's predictions are reliable and helps in improving the model through continuous evaluation

3.2 Non-Functional Requirements

3.2.1 Accuracy Module

The system must ensure high accuracy in both phenotype prediction and genotype inference. The models should be properly trained and validated to minimize errors and ensure reliable outputs. Accuracy is critical because incorrect predictions can lead to misleading biological interpretations.

3.2.2 Scalability Module

The system should be capable of handling large datasets, including thousands of rice leaf images and genomic records. It should efficiently manage increased data volume without significant degradation in performance.

3.2.3 Performance Module

The system should provide efficient processing at all stages, including preprocessing, feature extraction, and model inference. Response time should be optimized, especially during prediction.

3.2.4 Usability Module

The system should have a user-friendly interface that allows easy interaction. Users should be able to upload images and view results without requiring technical expertise.

3.2.5 Reliability Module

The system should produce consistent and stable results for similar inputs. It must handle variations in image quality and ensure robustness against noise and distortions.

3.2.6 Maintainability Module

The system should be designed in a modular manner so that updates can be made easily. New datasets or improvements in models should be integrated without major restructuring.

3.2.7 Flexibility Module

The system should support integration of new datasets and models. It should be adaptable to evolving technologies and future research developments.

3.2.8 Interpretability Module

The system should present outputs in an understandable manner. Even though the internal models may be complex, the final results should be clear and easy to interpret.

3.2.9 Security Module

The system should ensure safe handling of user data, including uploaded images. Proper data protection measures should be implemented

3.3 Constraints

The system has the following limitations:

- **Lack of direct genotype labels:** Most datasets contain images but not exact genotype mapping
- **Biological complexity:** Same phenotype may correspond to different genotypes due to environmental effects
- **Image variability:** Differences in lighting, background, and quality may affect feature extraction
- **Data integration challenge:** Combining image data with genomic datasets (SRA, pan-genome) is complex
- **Model interpretability:** Deep learning models may act as black boxes
- **Computation cost:** Training deep learning models requires high computational resources

These constraints were also identified in your PoC document under system limitations .

3.4 Hardware & Software Requirements

Table 2: Hardware Requirements

Component	Minimum	Recommended
Processor	Intel i5 / Ryzen 5	Intel i7 / Ryzen 7
RAM	8 GB	16–32 GB
GPU	Optional	NVIDIA GPU (RTX series, CUDA-enabled)
Storage	50 GB SSD	100 GB SSD

These are required for handling image data and training deep learning models .

Software Requirements :

- **Programming Language:** Python
- **Frameworks:** TensorFlow / PyTorch
- **Libraries:**
    - OpenCV (image processing)

    - NumPy, Pandas (data handling)
    - Matplotlib / Seaborn (visualization)
- **Tools:**
    - Jupyter Notebook (experiments)
    - Git (version control)
- **GPU Support:** CUDA, cuDNN

3.5 Techniques Utilized Till Date for the Proposed System

The development of the proposed system involves the integration of multiple techniques from image processing, machine learning, and genomics to enable effective phenotype-to-genotype inference.

1. Image Processing Techniques

Image processing forms the initial stage of the system, where raw leaf images are prepared for analysis. Techniques such as image resizing and normalization are applied to ensure uniform input dimensions and consistent pixel distributions. Background removal is performed to isolate the leaf region and eliminate noise caused by surrounding elements. To enhance the visibility of important structures, contrast enhancement techniques like CLAHE (Contrast Limited Adaptive Histogram Equalization) are used. Additionally, venation enhancement is carried out using vesselness filtering, which highlights the vascular structure of the leaf, making it easier to extract biologically relevant features.

2. Feature Extraction Techniques

Once the images are preprocessed, the system extracts meaningful features that represent the leaf phenotype. This includes venation-based features such as vein density, branching patterns, and network structure, which are closely linked to genetic traits. Along with this, texture features are extracted to capture variations in surface patterns, and geometric features are used to describe the shape and morphology of the leaf. These feature extraction techniques are inspired by existing research that emphasizes the importance of venation and structural characteristics in plant analysis.

3. Deep Learning Techniques

Deep learning methods are used to automatically learn complex patterns from the extracted features. The system explores Convolutional Neural Network (CNN)-based architectures such as ResNet and EfficientNet, which are effective in capturing hierarchical image features. A multi-branch architecture is also considered, where different types of features (venation, texture, and geometry) are processed in parallel. Furthermore, initial work has been carried out on feature fusion mechanisms, which combine multiple feature representations into a single unified form to improve prediction accuracy.

4. Genomic Data Utilization

To establish the connection between phenotype and genotype, the system incorporates genomic data from reliable biological databases. Datasets from sources such as NCBI SRA and rice genome databases like RGAP and SNP-Seek are explored. The system also studies large-scale genomic research, including pan-genome analysis and multi-variety genomic datasets, to understand genetic diversity and variation across rice populations. This integration enables the system to map phenotypic features to genotype-related information.

5. Conceptual Techniques

At the conceptual level, the system is based on the idea of linking phenotype and genotype using data-driven approaches. A GWAS (Genome-Wide Association Study)-based concept is considered to establish associations between observed traits and genetic markers. Additionally, there is a plan to integrate ontology-based structuring and metadata organization, as suggested in expert discussions (MoM), to improve data interpretation and scalability of the system.

3.6 Tools utilized till date for the proposed system

1. Programming Tools

- Python
- Jupyter Notebook
- Google Colab

2. Deep Learning Frameworks

- TensorFlow
- PyTorch

3. Image Processing Tools

- OpenCV
- NumPy
- Pandas

4. Visualization Tools

- Matplotlib

5. UI/Design Tools

- Figma

6. Version Control

- Git
- GitHub

7. Biological Databases

- NCBI SRA
- SNP-Seek
- RGAP

3.7 SDGs mapped with the proposed project and justification

SDG 2: Zero Hunger (Supporting SDG)

Justification:
Rice is a staple food crop, and improving its quality and yield is directly linked to food security. The proposed system helps in identifying and analyzing rice varieties based on phenotypic and genotypic characteristics, which can support better crop selection, breeding, and management practices. By enabling efficient analysis of plant traits, the system can contribute to enhanced agricultural productivity and sustainable food production.

SDG 4: Quality Education (Supporting SDG)

Justification:
The project promotes interdisciplinary learning by integrating concepts from computer science, artificial intelligence, and plant biology. It encourages research-based learning and practical application of theoretical knowledge, contributing to skill development and knowledge enhancement in emerging technological domains

SDG 9: Industry, Innovation and Infrastructure (Primary SDG)

Justification:
The proposed project strongly aligns with SDG 9 as it focuses on developing an innovative technological solution for agriculture using advanced methods such as deep learning, image processing, and genomic data analysis. The system introduces a novel approach for genotype inference from rice leaf phenotypes, which combines computer vision with biological data. This promotes innovation in the agricultural domain by replacing traditional, laboratory-based genetic analysis with a scalable, non-invasive digital solution. Furthermore, the integration of AI models and large genomic datasets contributes to the development of modern digital infrastructure for smart agriculture and precision farming.

SDG 13: Climate Action (Supporting SDG)

Justification:
The project supports climate action by enabling the identification of rice varieties with desirable traits such as resilience to environmental stress conditions. Understanding genotype–phenotype relationships can help in selecting crops that are better adapted to changing climatic conditions. This contributes to the development of climate-resilient agricultural systems.

.

4. Proposed Design

4.1 Block diagram representation of the proposed system with explanation

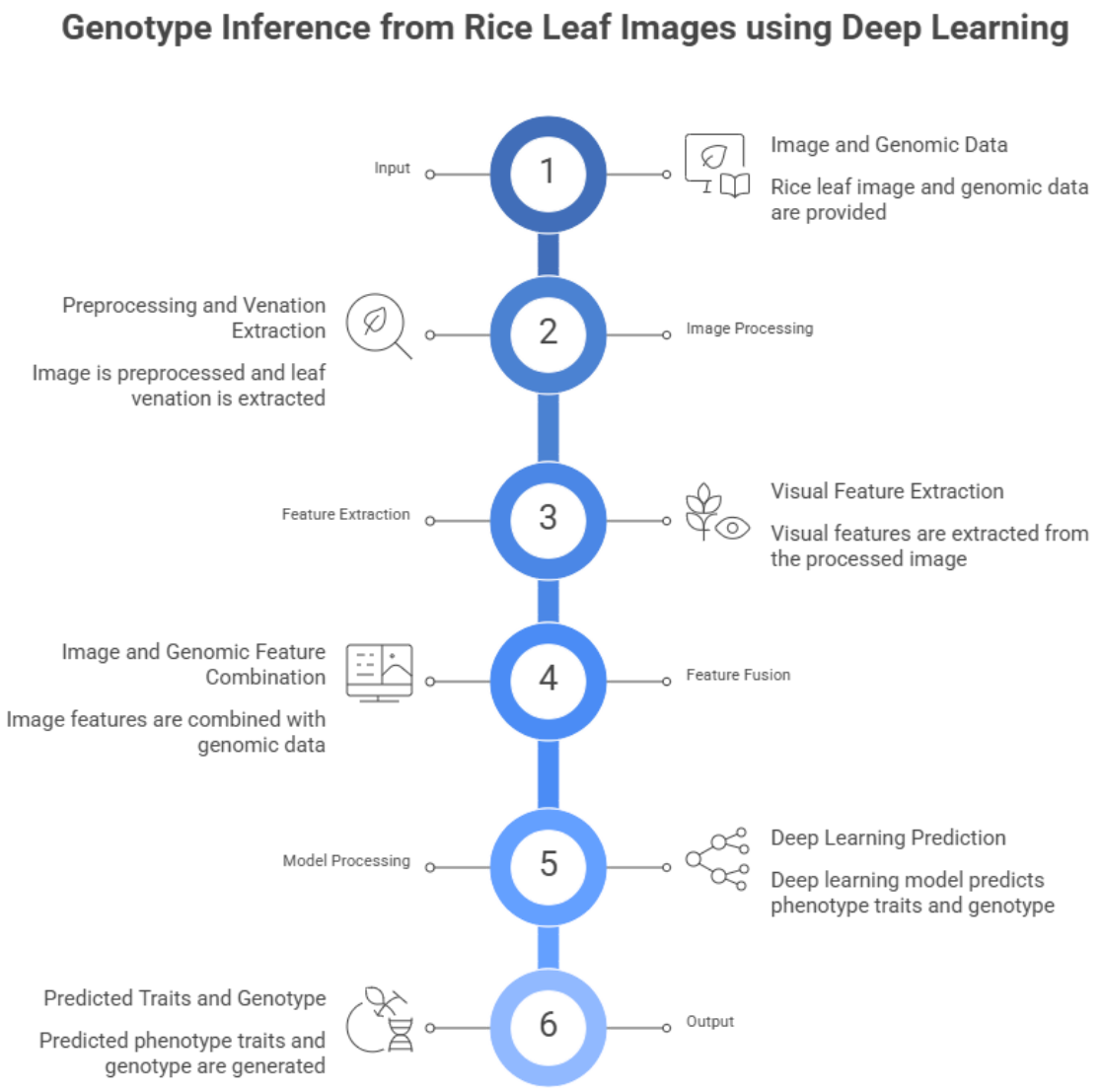


Figure 1: Workflow of genotype inference from rice leaf images using deep learning.

Diagram shows pipeline from input rice leaf images and genomic data, preprocessing, venation extraction, feature extraction, feature fusion, deep learning prediction, to final output of phenotype traits and genotype inference.

4.2 Modular diagram representation of the proposed system with explanation

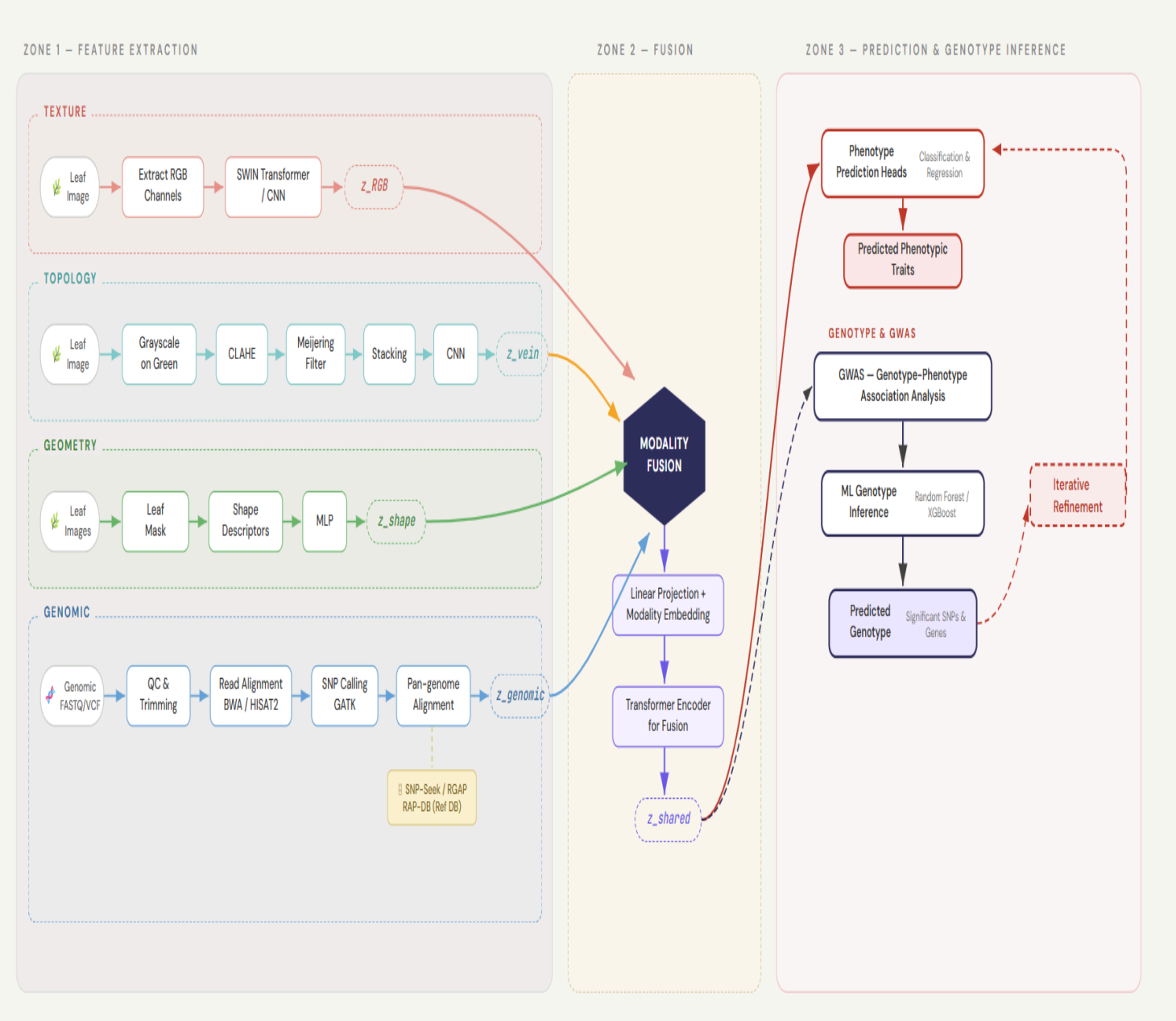


Figure 2: Multimodal Framework for Leaf Phenotype Prediction and Genotype Inference

The system integrates texture, venation, geometry, and genomic features using parallel branches, fuses them via a transformer, and performs phenotype prediction followed by genotype inference using GWAS-inspired methods.

4.3 Design of the proposed system –Use Case Diagram

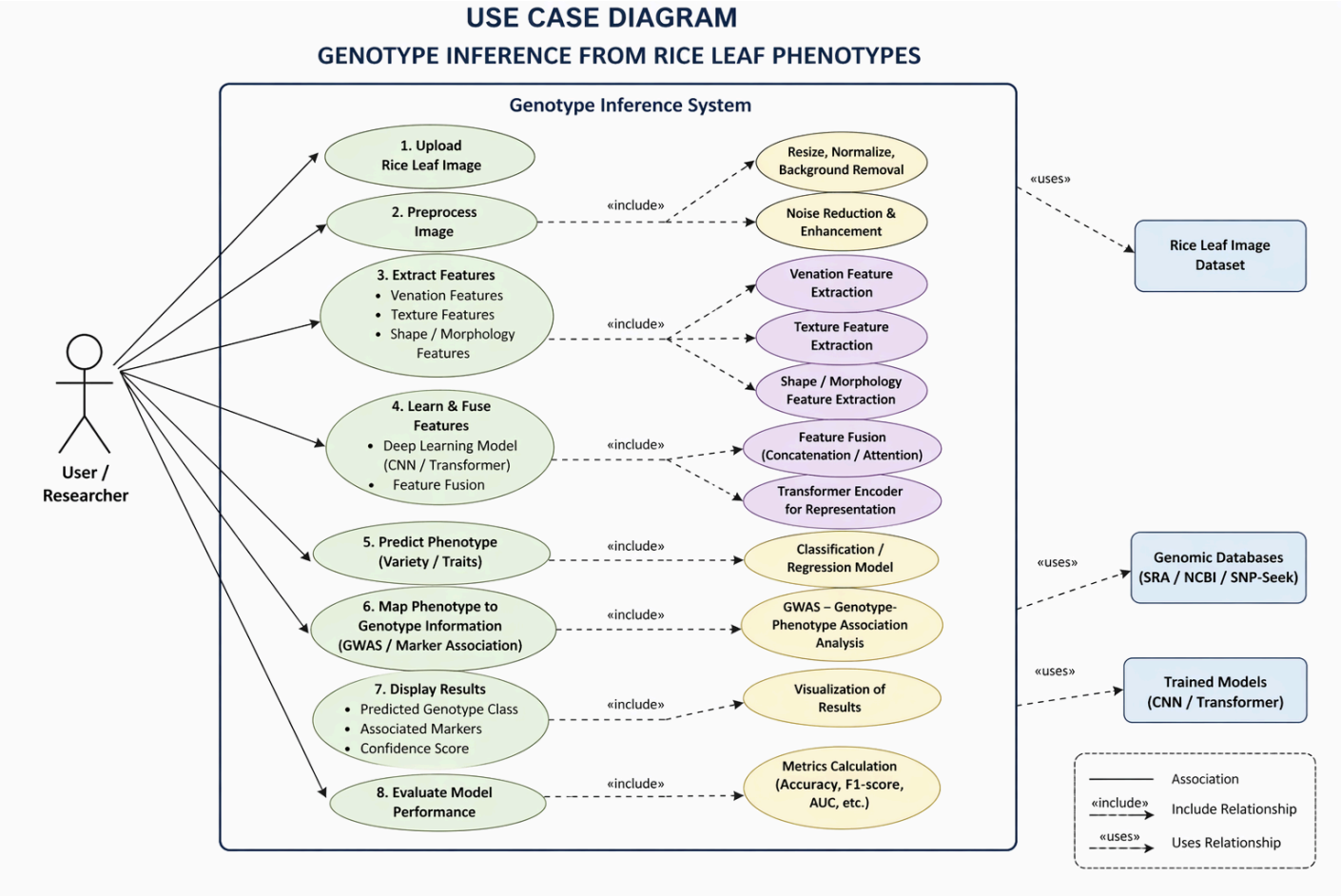


Figure 3: Use Case Diagram for Rice Leaf Phenotype-Based Genotype Inference System.

The diagram shows researcher interaction with the system, including image upload, preprocessing, feature extraction, model-based prediction, and mapping phenotypes to genotypes using GWAS methods and genomic databases.

4.4 Algorithms Utilized in Existing Systems

Various algorithms have been used in existing systems for plant analysis, leaf feature extraction, and genomic studies. These can be broadly categorized into image-based and genomic approaches.

1. Traditional Image Processing Algorithms

Edge Detection (Canny, Sobel):

Used to detect leaf boundaries and vein structures.

How it works:

- **Sobel** computes image gradients to highlight edges.
- **Canny** uses noise reduction, gradient calculation, non-maximum suppression, and edge tracking.

  Citation:
- Digital Image Processing — Rafael C. Gonzalez & Richard E. Woods

  Link:

  https://www.pearson.com/en-us/subject-catalog/p/digital-image-processing/P200000003354

Thresholding Techniques:

Used for segmentation of leaf regions from the background.

How it works:

- Converts grayscale image into binary using a threshold value.

  Citation:
- Digital Image Processing

  Link: https://books.google.com/books?id=Rz4lDwAAQBAJ

Morphological Operations:

Used to refine extracted vein structures.

How it works:

- Uses erosion and dilation to remove noise and enhance structures.

  Citation:

- Digital Image Processing
  Link:
  https://www.pearson.com/en-us/subject-catalog/p/digital-image-processing/P200000003354

2. Machine Learning Algorithms

Support Vector Machine (SVM):

Used for classification of plant species based on extracted features.

How it works:

- Finds a maximum-margin hyperplane to separate classes.
  Citation:
- Pattern Recognition and Machine Learning — Christopher M. Bishop
  Link: https://link.springer.com/book/10.1007/978-0-387-45528-0

K-Nearest Neighbors (KNN):

Used for similarity-based classification.

How it works:

- Assigns class based on nearest neighbors using distance metrics.
  Citation:
- Pattern Recognition and Machine Learning
  Link: https://books.google.com/books?id=kV1jDwAAQBAJ

Decision Trees:

Used for rule-based classification of plant features.

How it works:

- Splits dataset into branches based on feature conditions.
  Citation:
- Machine Learning — Tom M. Mitchell
  Link: https://www.cs.cmu.edu/~tom/mlbook.html

3. Deep Learning Algorithms

Convolutional Neural Networks (CNN):

Widely used for automatic feature extraction from leaf images.

How it works:

- Uses convolution, pooling, and fully connected layers for classification.

  Citation:

- Deep Learning

  Link: https://www.deeplearningbook.org/

ResNet / EfficientNet:

Used for improved accuracy and deeper feature learning.

How it works:

- **ResNet:** Uses skip connections to solve vanishing gradient problems.
- **EfficientNet:** Scales network dimensions efficiently.

  Citation:

- Deep Learning

  Link: https://www.deeplearningbook.org/

Vision Transformers (ViT / Swin Transformer):

Used for capturing global patterns in images.

How it works:

- Splits images into patches and processes using attention mechanisms.

  Citation:

- Deep Learning

  Link: https://www.deeplearningbook.org/

4. Segmentation Algorithms

U-Net / DeepLabV3+:

Used for precise leaf vein segmentation.

How it works:

- U-Net uses encoder-decoder with skip connections.

- DeepLabV3+ uses atrous convolution and spatial pyramid pooling.

  Citation:
- Deep Learning for Computer Vision — Rajalingappaa Shanmugamani

  Link:

  https://www.packtpub.com/product/deep-learning-for-computer-vision/9781788295628

5. Genomic Analysis Algorithms

Sequence Alignment (BWA, HISAT2):

Used to align genomic sequences.

How it works:

- Uses indexing techniques like Burrows-Wheeler Transform.

  Citation:
- Bioinformatics Algorithms

  Link: https://www.bioinformaticsalgorithms.org/

Variant Calling (GATK):

Used to identify SNPs and genetic variations.

How it works:

- Detects genomic differences using statistical models.

  Citation:
- Bioinformatics Algorithms

  Link: https://www.bioinformaticsalgorithms.org/

GWAS (Genome-Wide Association Study):

Used to associate phenotypic traits with genetic markers.

How it works:

- Identifies correlations between SNPs and traits across genomes.

  Citation:
- Introduction to Genetic Analysis

  Link: https://www.ncbi.nlm.nih.gov/books/NBK21134/

Whole Genome Sequencing (WGS):

Used to analyze the complete DNA sequence of an organism's genome.

How it works:

- The entire genome is fragmented, sequenced into small reads, and then computationally assembled into a complete sequence.
- Enables identification of all genetic variations, including SNPs, insertions, deletions, and structural variants.

  Citation:
- Introduction to Bioinformatics — Arthur M. Lesk

  Link:

  https://global.oup.com/academic/product/introduction-to-bioinformatics-9780198794141

4.5 Project scheduling and tracking using Timeline / Gantt Chart



Figure 4: Project Scheduling and Tracking Using Gantt Chart

The chart shows the complete project timeline across 7 phases (Jan–Aug 2026), covering literature review, dataset collection, preprocessing, venation extraction, deep learning model development, genotype mapping using GWAS methods, and final evaluation.

5. Implementation Results and Discussions

5.1. Data Collection / Acquisition

The system ingests data from two primary source types — visual imagery of leaf specimens and raw genomic sequencing files — which together constitute the multimodal input to the pipeline.

5.1.1. Image Data

Leaf images are acquired using standard RGB digital cameras or scanners under controlled illumination conditions. Each image captures the full lamina of the leaf specimen and is stored in a lossless or high-quality compressed format (PNG/JPEG). To ensure uniformity, images are standardised to 256 × 256 pixels during the preprocessing stage. Three independent visual modalities are derived from the imagery:

- Texture: RGB channel data conveying surface colouration, pigmentation, and disease spotting patterns.
- Topology: The green channel, specifically processed to expose vein network architecture.
- Geometry: Binary leaf masks extracted to compute morphometric shape descriptors.

5.1.2. Genomic Data

Genomic input is sourced as raw sequencing reads (FASTQ format) or pre-called variant files (VCF format). These files are produced by next-generation sequencing (NGS) platforms targeting the organism's nuclear genome. Reference databases used during alignment and SNP annotation include:

- SNP-Seek: Rice SNP and haplotype database.
- RGAP (Rice Genome Annotation Project): Structural and functional gene annotations.
- RAP-DB (Rice Annotation Project Database): High-quality rice genome reference.

5.2. Data Preprocessing Techniques

Table 3: Feature Extraction runs four parallel modality tracks:

Track	**Core Operations**	**Output**
Texture	Load → Resize → CLAHE (LAB) → GLCM	z_RGB

Topology	Green channel → CLAHE → Top-Hat → Adaptive threshold → Skeletonize	z_vein
Geometry	Otsu mask → Contour → 6 shape descriptors → MLP	z_shape
Genomic	FASTQ/VCF → QC → BWA/HISAT2 → GATK SNP → Pan-genome	z_genomic

Texture Track — Detailed Steps

The texture modality captures surface-level spectral and statistical properties of the leaf:

Image Loading & Resizing: Raw BGR images are loaded using OpenCV and resized to a fixed resolution of 256×256 pixels to ensure consistent input dimensionality.

- CLAHE Enhancement: The image is converted to LAB colour space. Contrast Limited Adaptive Histogram Equalisation (CLAHE, clipLimit=3.0) is applied exclusively to the luminance (L) channel to enhance local contrast without distorting chrominance information.
- GLCM Feature Extraction: A Gray-Level Co-occurrence Matrix (GLCM) is computed at a displacement of [1] and angle of [0°] on the grayscale image. Three Haralick statistics are extracted: Contrast (local intensity variation), Homogeneity (closeness of element distribution to the diagonal), and Energy (sum of squared elements — uniformity measure).
- Deep Encoding: The enhanced RGB image is passed through a SWIN Transformer or CNN to produce the latent texture vector z_RGB.

Topology Track — Detailed Steps

The topology modality recovers the leaf vein network structure:

- Green Channel Isolation: The green channel (index 1) is extracted from the BGR image, as vein structures exhibit the highest contrast in this band due to chlorophyll absorption properties.
- CLAHE: Tile-based CLAHE (clipLimit=3.0, tileGridSize=(8,8)) is applied to the green channel to equalise contrast across spatial regions.
- Gaussian Blur: A 5×5 Gaussian blur smooths noise prior to morphological filtering.

- Morphological Top-Hat Transform: An elliptical structuring element (15×15 px) is used in a White Top-Hat transform to isolate bright thin structures (veins) from the darker leaf background. This corresponds functionally to the Meijering neurite filter step in the architecture diagram.
- Adaptive Thresholding: Gaussian adaptive thresholding (block size=11, C=2) binarises the top-hat response. The resulting vein map is AND-masked with the leaf boundary mask to suppress background noise.
- Skeletonization: The binary vein map is morphologically thinned to a one-pixel-wide skeleton. Vein density (skeleton pixel count / mask area) and branch-point count (convolution-based detection of pixels with ≥3 neighbours) are computed as topology features.
- CNN Encoding: The stacked vein feature maps are encoded by a CNN to produce z_vein.

Geometry Track — Detailed Steps

The geometry modality quantifies the macroscopic shape of the leaf:

- Leaf Segmentation: Otsu's global threshold is applied to the grayscale image to produce a binary mask. Small connected components (< 500 pixels) are removed to eliminate artefacts.
- Contour Analysis: The dominant external contour is extracted using cv2.findContours. Six shape descriptors are computed: (1) Area — total leaf pixel area; (2) Perimeter — arc length of the contour; (3) Aspect Ratio — width/height of the bounding rectangle; (4) Circularity — $4\pi A/P^2$, measuring compactness; (5) Solidity — ratio of contour area to convex hull area, sensitive to leaf serrations; (6) Eccentricity — derived from the minor/major axes of a fitted ellipse.
- MLP Encoding: The six-dimensional feature vector is projected through a multi-layer perceptron to produce z_shape.

Genomic Track — Detailed Steps

The genomic modality captures molecular-level genetic variation:

- QC & Trimming: Raw FASTQ reads are quality-checked and adapter sequences / low-quality bases are trimmed.

- Read Alignment: Trimmed reads are aligned to the reference genome using BWA-MEM (for DNA) or HISAT2 (for splicing-aware RNA alignment). Aligned reads are stored as sorted BAM files.
- SNP Calling: GATK HaplotypeCaller performs variant calling on the aligned BAM to produce a VCF file containing SNPs and small INDELs.
- Pan-genome Alignment: Variants are cross-referenced against SNP-Seek, RGAP, and RAP-DB reference databases to annotate allele frequencies and associated gene loci. The resulting variant profile is encoded as z_genomic.

5.3. Experimental Setup

Table 4: Phases of Experiments

Phase	**Activity**	**Duration**
Phase 1	Problem Identification & Literature Survey	Jan – Feb
Phase 2	Dataset Collection & Analysis	Feb – March
Phase 3	System Design & Architecture	March
Phase 4	Preprocessing & Feature Extraction	April
Phase 5	Model Development	May
Phase 6	Genotype Mapping & Integration	June
Phase 7	Testing & Evaluation	July
Phase 8	Documentation & Final Submission	Aug

5.4 Evaluation Metrics

To evaluate the performance of the proposed system, both machine learning metrics and domain-specific venation metrics are considered.

1. Accuracy

Measures the overall correctness of predictions made by the model.

Accuracy = (Correct Predictions / Total Predictions)

It helps in understanding how well the model predicts rice varieties or genotype-associated classes.

2. Precision, Recall, and F1-Score

These are used for more detailed evaluation:

- **Precision:** Measures how many predicted positives are actually correct
- **Recall:** Measures how many actual positives are correctly identified
- **F1-Score:** Balance between precision and recall

These metrics are important when dealing with multiple rice varieties or imbalanced datasets.

3. Loss Function

- **Categorical Cross-Entropy** is used for classification tasks
- **Mean Squared Error (MSE)** may be used for regression outputs

Loss helps in monitoring how well the model is learning during training.

4. Venation-Specific Metrics

Since the project focuses on leaf venation, special metrics are used:

- **Edge Density:**

  Measures the proportion of edges in the image. Helps detect noise and unwanted features.
- **Vein Coverage:**

  Indicates how much of the leaf area is identified as venation.
- **Skeleton Length:**

  Measures continuity of the vein network.
- **Branch Points:**

  Indicates complexity of the venation structure.
- **Entropy:**

  Measures the amount of information and randomness in the extracted venation pattern.

These metrics are important because they directly evaluate how well the system captures biologically relevant features.

5. Model Validation

- Train-test split and cross-validation are used
- Performance is compared with baseline models and existing approaches

5.5 Gap Analysis of Existing Systems

Even though significant work has been done in plant analysis and rice genomics, there are still several gaps that this project aims to address.

1. Separation of Phenotype and Genotype Studies

Most genomic studies focus only on DNA-level analysis, such as pan-genome studies and SNP analysis in rice datasets.

On the other hand, image-based systems focus only on visual classification.There is very limited work that connects both domains.

2. Limited Use of Venation Patterns

Many existing image-based models rely on:

- Color
- Texture

However, venation patterns, which are more genetically influenced, are often not fully utilized.

3. Focus on Classification, Not Inference

Most systems:

- Identify plant species
- Detect diseases

But they do not attempt genotype inference, which is a deeper and more complex problem.

4. Lack of Integrated Datasets

- Image datasets and genomic datasets are usually separate
- Very few systems combine:

    - Leaf images
    - Genomic data (SRA, SNP-Seek)

This creates a major gap in building a complete system.

5. Scalability Issues in Traditional Methods

Traditional genotyping methods:

- Require lab setup
- Are not suitable for large-scale field use

There is a need for a fast and scalable alternative, which this project attempts to provide.

6. Interpretability Challenges

Even in advanced models:

- Results are not always biologically explainable
- There is limited understanding of how features relate to genetic traits

6. Plan of Action for the Next Semester

6.1 Work Done Till Date (Project I – Semester VI)

During the first phase of the project, the team focused on understanding the problem, exploring feasible solutions, and building a strong foundation.

Initially, the project idea was based on medicinal plants, but after discussions with experts and availability of datasets, the direction was refined towards rice plants, as they have well-structured genomic data and better research support. This shift made the project more practical and data-driven.

A detailed literature survey was carried out to understand existing work in areas such as leaf venation analysis, deep learning-based plant classification, and rice genomics. Studies involving large-scale rice genome datasets and pan-genome analysis helped in understanding the complexity and diversity of rice genetics.

The team also explored various biological databases such as NCBI SRA, SNP-Seek, and rice genome annotation platforms to understand how genotype data can be accessed and used.

A complete system architecture and methodology was designed. This includes:

- Image preprocessing and venation enhancement
- Multi-feature extraction (venation, texture, shape)
- Deep learning models for feature learning
- Conceptual mapping of phenotype to genotype

Initial work was also done on:

- Dataset understanding and preprocessing
- Figma-based UI design
- Exploring model architectures and feature extraction techniques

Additionally, a meeting with domain experts at ACTREC helped in refining the project scope, emphasizing the use of real genomic datasets, ontology-based structuring, and data integration techniques.

6.1 Plan of Action for Project II

In the next semester, the focus will shift from planning and design to implementation, integration, and validation of the system.

1. Dataset Preparation and Integration

- Collect and organize:
    - Rice leaf image datasets
    - Genomic datasets from SRA and SNP-Seek
- Perform data cleaning, labeling, and alignment between phenotype and genotype data
- Build a structured dataset for training and testing

2. Database Development

- Develop a **Rice Genome Database** including:
    - Genomic sequences
    - SNP data
    - Metadata and annotations
- Apply structured organization techniques (ontology-based approach as discussed in MoM)

3. Model Development

- Implement:
    - Image preprocessing pipeline
    - Venation extraction and enhancement
- Train deep learning models (CNN/Transformer) for feature extraction
- Develop feature fusion mechanism for combining different features

4. Phenotype–Genotype Mapping

- Apply GWAS-inspired methods to link phenotypic traits with genotype data
- Use machine learning models (Random Forest / XGBoost) for genotype inference

5. System Integration

- Combine all modules into a complete pipeline:
    - Input → Processing → Prediction → Output
- Develop a simple user interface for interaction

6. Evaluation and Optimization

- Evaluate model using:
    - Accuracy, precision, F1-score
    - Venation-specific metrics
- Optimize model performance and improve robustness

7. Testing and Validation

- Perform testing on different datasets
- Validate results using existing biological research findings

8. Documentation and Final Presentation

- Prepare final report and documentation
- Create presentation and demo
- Prepare research paper based on results

7. Conclusion

The project focuses on developing a novel approach to understanding the relationship between phenotype and genotype in rice plants by using leaf venation patterns and deep learning techniques. Instead of relying on traditional laboratory-based genetic analysis, which is often time-consuming and resource-intensive, the proposed system explores the possibility of extracting meaningful biological insights directly from leaf images. Through the course of the project, the work has evolved from an initial idea based on medicinal plants to a more structured and data-supported study on rice. This shift allowed the use of well-established genomic resources such as the NCBI SRA and SNP-Seek, which provide large-scale genomic and variation data. These datasets form an important foundation for linking visual traits with genetic information. The system designed in this project integrates multiple components, including image preprocessing, venation feature extraction, deep learning-based pattern recognition, and genotype association using machine learning models. By combining these elements, the project attempts to bridge a key gap between image-based plant analysis and genomic research, which are traditionally treated as separate domains. Although the relationship between phenotype and genotype is complex and influenced by multiple factors, this work demonstrates a structured approach toward modeling this connection in a computational way. The use of venation patterns, which are relatively stable and biologically significant, adds an important dimension to the analysis compared to conventional image-based methods that rely mainly on surface-level features. In its current stage, the project has established a strong conceptual and methodological foundation. The next phase will focus on full implementation, model optimization, and validation using real-world datasets. If successfully developed, the system can provide a scalable and non-invasive tool for applications in agriculture, such as crop monitoring, variety identification, and support for breeding programs.

Overall, the project contributes toward the growing intersection of Computer Vision, Deep Learning, and Plant Genomics, offering a data-driven approach to understanding plant traits and their genetic basis in an efficient and practical manner.

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9. Appendix

Table-I: List of Figures

Fig No.	Heading	Pg. no.
1.	Workflow of genotype inference from rice leaf images using deep learning.	21
2.	Multimodal Framework for Leaf Phenotype Prediction and Genotype Inference	22
3.	Use Case Diagram for Rice Leaf Phenotype-Based Genotype Inference System.	23
4.	Project scheduling and tracking using Timeline / Gantt Chart	37

Table-II: List of tables

Table Number	Heading	Page no.
1	Dataset Collected	11-12
2	Literature Review	14-17
2	Hardware Requirements	24
3	Feature Extraction	38
4	Phases of Experiments	41

General Instructions:

Project Review 1 Evaluation sheet:

Industry / Inhouse:

Research / Product Innovation / Application/ Others **Capstone Project Evaluation Sheet 2025-26**

SDG No.: 3 (27)

Class: D12A/B/C

Title of Project (Group no) 27 Genotype Inference from Leaf Phenotype Using Venation Patterns & Deep Learning

Mentor Name & Group Members: Dr. Priya R.L. (46) Meghana Poojary, Shivam Chaugule (10), Yash Sharma, Rama Yadav (56) (64)

	Engineering Concepts & Knowledge (5)	Interpretation of Problem & Analysis (3)	Sustainable Goal Mapped (2)	Design / Prototype (5)	Interpretation of Data & Dataset (3)	Modern Tool Usage (5)	Societal Benefit, Safety Consideration (2)	Environment Friendly & Cost Effective (2)	Ethics (2)	Team work (2)	Presentation Skills (3)	Applied Engg & Mgmt Standards (3)	Life-long learning (3)	Professional Skills (5)	Innovative / Research Approach (5)	Total Marks (50)
Review of Project Stage 1	4	3	2	4	2	4	2	2	2	2	2	3	3	4	4	43

Comments:

Priya R.L

Name & Signature Reviewer1

	Engineering Concepts & Knowledge (5)	Interpretation of Problem & Analysis (3)	Sustainable Goal Mapped (2)	Design / Prototype (5)	Interpretation of Data & Dataset (3)	Modern Tool Usage (5)	Societal Benefit, Safety Consideration (2)	Environment Friendly & Cost Effective (2)	Ethics (2)	Team work (2)	Presentation Skills (3)	Applied Engg & Mgmt Standards (3)	Life-long learning (3)	Professional Skills (5)	Innovative / Research Approach (5)	Total Marks (50)
Review of Project Stage 1	4	3	2	4	1	4	2	2	2	2	2	3	3	4	4	42

Comments: Good literature Review 1) Dataset Preparation should be stated earlier.
2) Parameters/features identified and shown.

Date: 21/02/2026

Yugchhaya Galphat (Mrs.)

Name & Signature Reviewer2

Project Review 2 Evaluation sheet:

Industry / Inhouse: Tata Memorial Centre (ACTREC), Khargar
Research / Product Innovation / Application/ Others **Capstone Project Evaluation Sheet 2025-26**
SDG No.: 9
Class: D12A/B/C

Title of Project (Group no): 27 Genotype inference from Phenotype pattern of leaf veins

Mentor Name & Group Members: Dr. Biju R. Meghana Bojagi (46), Shivam Chaugule (10), Yash Sharma (52), Rama Yadav (64)

	Engineering Concepts & Knowledge (5)	Interpretation of Problem & Analysis (3)	Sustainable Goal Mapped (2)	Design / Prototype (5)	Interpretation of Data & Dataset (3)	Modern Tool Usage (5)	Societal Benefit, Safety Consideration (2)	Environment Friendly & Cost Effective (2)	Ethics (2)	Team work (2)	Presentation Skills (3)	Applied Engg & Mgmt Standards (3)	Life-long learning (3)	Professional Skills (5)	Innovative / Research Approach (5)	Total Marks (50)
Review of Project Stage 1	5	4	2	4	2	4	2	2	2	2	3	2	2	4	3	44

Comments:
→ weight of different components needs to be validated with different Model or Algorithm
→ Data set needs to be explored in detail

Mrs. Jyoti Chaure G. Chaure
Name & Signature Reviewer1

	Engineering Concepts & Knowledge (5)	Interpretation of Problem & Analysis (3)	Sustainable Goal Mapped (2)	Design / Prototype (5)	Interpretation of Data & Dataset (3)	Modern Tool Usage (5)	Societal Benefit, Safety Consideration (2)	Environment Friendly & Cost Effective (2)	Ethics (2)	Team work (2)	Presentation Skills (3)	Applied Engg & Mgmt Standards (3)	Life-long learning (3)	Professional Skills (5)	Innovative / Research Approach (5)	Total Marks (50)
Review of Project Stage 1	5	3	2	4	2	4	2	2	2	2	3	3	3	4	4	45

Comments:
Dataset extraction to be started. Identify every features and its corresponding weightage for the case.

Date: 25/03/2026

Name & Signature Reviewer2